

Cellphone-Based Device Control with Voice Acknowledgement

■ AZARUDEEN ANIFA

Here is a circuit that lets you operate your home appliances like lights and water pump from your office or any other remote place. So if you forgot to switch off the lights or other appliances while going out, it helps you to turn off the appliance with your cellphone. Your cellphone works as the remote control for your home appliances. You can control the desired appliance by pressing the corresponding key. The system also gives you voice acknowledgement of the appliance status.

Circuit description

Fig. 1 shows the circuit for cellphone-based device control with voice acknowledgement. It comprises microcontroller AT89C51, DTMF decoder MT8870, voice recording/playback device APR9600 and a few discrete components.

Microcontroller AT89C51 is at the heart of the circuit. It is a low-power, high-performance, 8-bit microcontroller with 4 kB of flash programmable and erasable read-only memory (PEROM) used as on-chip program memory, 128 bytes of RAM used as internal data memory, 32 individually programmable input/output (I/O) lines divided into four 8-bit ports, two 16-bit programmable timers/counters, a five-vector two-level interrupt architecture, on-chip oscillator and clock circuitry. A 11.0592MHz crystal (X_{TAL1}) is used to provide basic clock frequency for the microcontroller. Capacitor C3 and resistor R3 form the power-on reset circuit, while push-to-on switch S20 is used for manual reset.

Port pins P1.0 through P1.7 of the microcontroller are configured to get

the input from push-to-on switches S1 through S8. Pins of Port P1 are pulled high via resistor network RNW1. Port pins P2.0 through P2.4 are configured to receive the decoded DTMF signal from DTMF receiver MT8870. The functions of the corresponding switches (S1 through S8) and cellphone keys are shown in Table 1.

The DTMF decoder is used for de-

TABLE I
Functions of Switches and Mobile Keys

Switches	Mobile key	Function
S1	1	Select Device 1; On/Off
S2	2	Select Device 2; On/Off
S3	3	Select Device 3; On/Off
S4	4	Select Device 4; On/Off
S5	5	Select Device 5; On/Off
S6	6	Select Device 6; On/Off
S7	*	Initialise to inform the status of the device
S8	#	On/off the voice alert

coding the mobile signal. It gets DTMF tone from the mobile headset's speaker pins and decodes it into 4-bit digital signal. The DTMF decoder is operated with a 3.579MHz crystal (X_{TAL2}). In DTMF receiver MT8870 (IC3), capacitor C12 is used to filter the noise and resistors R6 and R7 help to amplify the input signal using the internal amplifier.

Pin 16 of IC3 connected to resistor R5 provides the early steering output. It goes high immediately when the digital algorithm detects a valid tone pair (signal condition). Any momentary loss of signal condition causes EST to return to low state.

Pin 17 of IC3 connected to capacitor C11 is bidirectional, acting as steering input/guard time output (St/GT). A

voltage greater than threshold of the steering logic V_{TH} detected at St causes the device to register the detected tone pair. The guard time output resets the external steering time constant, and its state is a function of EST and the volt-



PARTS LIST

Semiconductors:

- IC1 - AT89C51 microcontroller
- IC2 - APR9600 voice recording and playback device
- IC3 - MT8870 DTMF receiver
- IC4 - ULN2003 relay driver
- IC5 - 7806, 6V regulator
- D1-D5 - 1N4007 diode
- LED1-LED3 - 5mm LED

Resistors (all 1/4-watt, ±5% carbon):

- R1-R3 - 10-kilo-ohm
- R4, R6, R15 - 100-kilo-ohm
- R5 - 470-kilo-ohm
- R7 - 220-kilo-ohm
- R8 - 39-kilo-ohm
- R9, R10 - 4.7-kilo-ohm
- R11 - 1-kilo-ohm
- R12 - 220-kilo-ohm
- R13, R14, R16 - 470-ohm
- RN1, RNW2 - 10-kilo-ohm resistor network

Capacitors:

- C1, C2 - 33pF ceramic disk
- C3 - 10µF, 16V electrolytic
- C4, C6-C8, C11, C15 - 0.1µF ceramic disk
- C5 - 4.7µF, 16V electrolytic
- C9, C10 - 22µF, 50V electrolytic
- C12 - 0.22µF polyester
- C13, C14 - 47µF polyester
- C16 - 1000µF, 35V electrolytic

Miscellaneous:

- S1-S16, S20 - Push-to-on switch
- S17, S18 - 2-way DIP switch
- S19 - SPDT switch
- S21 - On/off switch
- X_{TAL1} - 11.0592MHz crystal
- X_{TAL2} - 3.579MHz crystal
- RL1-RL6 - 6V, 1C/O relay
- MIC1 - Electret mic
- LS1 - 1W, 16-ohm speaker
- CON1 - Mobile phone headset connector